Options and Derivatives

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1 Question 1: Strangle Strategy

1.1 Payoff Structure

A long strangle consists of purchasing a European put with strike K_1 and a European call with strike K_2, where $K_1 < K_2$, sharing the same maturity T. Denoting the put and call premiums by p_0 and c_0, the payoff and profit at expiry are

$$\Pi_T = \max(K_1 - S_T, 0) + \max(S_T - K_2, 0), \qquad Y_T = \Pi_T - (p_0 + c_0).$$

Table 1: Long strangle payoff and profit by terminal stock price region

Region	Put payoff	Call payoff	Total payoff	Profit
$S_T \leq K_1$	$K_1 - S_T$	0	$K_1 - S_T$	$K_1 - S_T - (p_0 + c_0)$
$K_1 < S_T < K_2$	0	0	0	$-(p_0 + c_0)$
$S_T \geq K_2$	0	$S_T - K_2$	$S_T - K_2$	$S_T - K_2 - (p_0 + c_0)$

The maximum loss equals the total premium paid, realised when $K_1 < S_T < K_2$.

1.2 Profit Diagram

The break even prices are $S_T^- = K_1 - (p_0 + c_0)$ and $S_T^+ = K_2 + (p_0 + c_0)$. Between the strikes the profit is flat at $-(p_0 + c_0)$, outside this range it rises linearly, with unlimited upside and downside bounded by $S_T = 0$.

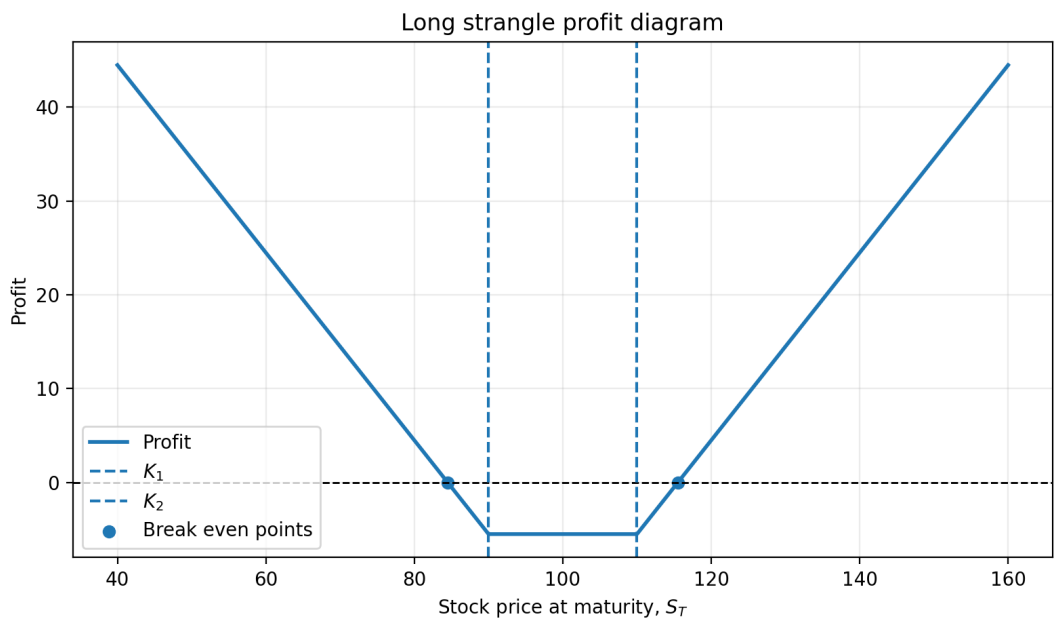

Figure 1: Illustrative profit diagram for a long strangle

1.3 Economic Interpretation

A long strangle is used when an investor expects high volatility but is uncertain about the direction of the price move. Compared with a straddle, the strategy has a lower initial cost because both options are out-of-the-money, but it requires a larger price movement to become profitable. The strategy is most attractive when implied volatility is low relative to the investor's expectation of realised volatility, since option premiums are then relatively cheap compared with the expected magnitude of the price move (Cui, 2026).

2 Question 2: Put–Call Parity Violation

2.1 Parity Test

For European options on a non-dividend paying stock, put–call parity requires $C_0 + Ke^{-rT} = P_0 + S_0$. Given $S_0 = 250$, $K = 235$, $C_0 = 21.88$, $P_0 = 5.25$, $r = 0.01$, and $T = 90/365$:

$$C_0 + Ke^{-rT} = 256.30 \neq 255.25 = P_0 + S_0.$$

The parity implied put price is $P_0^* = C_0 + Ke^{-rT} - S_0 = 6.30$, so the market put at \$5.25 is underpriced by \$1.05 (see Appendix B.1 for detailed calculations).

2.2 Arbitrage Strategy

Table 2: Arbitrage portfolio construction

Action	Cash flow at $t = 0$	Payoff at T
Buy stock	-250.00	S_T
Buy put	-5.25	$\max(K - S_T, 0)$
Sell call	$+21.88$	$-\max(S_T - K, 0)$
Borrow Ke^{-rT}	$+234.42$	$-K$
Net	**+1.05**	**0**

At maturity the combined position pays $S_T + \max(K - S_T, 0) - \max(S_T - K, 0) - K = 0$ in every state (verified for both $S_T \geq K$ and $S_T < K$ in Appendix B.1). The strategy yields a riskless profit of \$1.05 today with zero liability at expiry. In practice, transaction costs and bid–ask spreads may reduce the \$1.05 profit, explaining why such mispricings can persist briefly in real markets.

3 Question 3: Brownian Motion Expectation

Let $\{W_t\}_{t\geq 0}$ be a standard Wiener process. Define $Z := W_{t+2} - W_t \sim N(0, 2)$, independent of $\mathcal{F}_t$, so that $W_{t+2} = W_t + Z$. Expanding each power of $W_{t+2} = W_t + Z$ and applying $\mathbb{E}[Z] = 0$, $\mathbb{E}[Z^2] = 2$, $\mathbb{E}[Z^3] = 0$ (full expansion in Appendix B.2):

$$\mathbb{E}_t[W_{t+2}] = W_t, \qquad \mathbb{E}_t[W_{t+2}^2] = W_t^2 + 2, \qquad \mathbb{E}_t[W_{t+2}^3] = W_t^3 + 6W_t.$$

$$\boxed{\mathbb{E}_t\left[W_{t+2} + 2W_{t+2}^2 + 3W_{t+2}^3\right] = 3W_t^3 + 2W_t^2 + 19W_t + 4.}$$

4 Question 4: One-Period Binomial Model

Parameters. $S_0 = 110$, $u = 1.1$, $d = 0.9$, $R = 1.05$, $K = 100$, $S_u = 121$, $S_d = 99$.

4.1 Risk-Neutral Probability

$$q = \frac{R-d}{u-d} = \frac{1.05-0.9}{0.2} = 0.75.$$

4.2 European Call and Put Prices

Terminal payoffs: $C_u = 21$, $C_d = 0$; $P_u = 0$, $P_d = 1$.

$$C_0 = \frac{qC_u + (1-q)C_d}{R} = \frac{0.75(21)}{1.05} = 15, \qquad P_0 = \frac{0.25(1)}{1.05} \approx 0.2381.$$

4.3 Replicating Portfolios

Each option is replicated by Δ shares and a bond position B (derivation in Appendix B.3).

Put: $\Delta = -\frac{1}{22}$, $B = 5.2381$. Cost: $\Delta S_0 + B = 0.2381 = P_0$. ✓

Call: $\Delta = \frac{21}{22}$, $B = -90$. Cost: $\Delta S_0 + B = 15 = C_0$. ✓

4.4 Put–Call Parity Verification

$$C_0 + \frac{K}{R} = 15 + \frac{100}{1.05} = 110.2381 = 0.2381 + 110 = P_0 + S_0. \checkmark$$

5 Question 5: American Options

Parameters. $S_0 = 100$, $K = 100$, $u = 1.10$, $d = 0.90$, $r_{\Delta t} = 0.025$, $R = 1.025$, $q = 0.625$.

Stock lattice: $S_u = 110$, $S_d = 90$, $S_{uu} = 121$, $S_{ud} = 99$, $S_{dd} = 81$.

5.1 American Call Option

Terminal payoffs: $C_{uu} = 21$, $C_{ud} = C_{dd} = 0$. Backward induction yields $C_u = 12.80$, $C_d = 0$, and $C_0 = 7.808$ (Appendix B.4). Early exercise at node u gives $(110-100)^+ = 10 < 12.80$, it is therefore better to hold the option rather than exercise it. Hence $C_0^{\text{Am}} = C_0^{\text{Eu}} \approx 7.808$.

5.2 American Put Option

Terminal payoffs: $P_{uu} = 0$, $P_{ud} = 1$, $P_{dd} = 19$.

At node u: continuation value $= 0.366$, exercise value $= 0$. $\Rightarrow P_u^{\text{Am}} = 0.366$.

At node d: continuation value $= 7.561$, exercise value $= 10 > 7.561$. $\Rightarrow P_d^{\text{Am}} = 10$. **Early exercise is optimal.**

$$P_0^{\text{Eu}} = 2.989, \qquad P_0^{\text{Am}} = 3.882.$$

The early exercise premium is $P_0^{\text{Am}} - P_0^{\text{Eu}} = 0.893$, reflecting the value of being able to exercise the put at node d rather than waiting (full derivation in Appendix B.4).

5.3 Put–Call Parity

For European options: $C_0^{\text{Eu}} - P_0^{\text{Eu}} = 4.819 = S_0 - K/R^2$. ✓

For American options: $C_0^{\text{Am}} - P_0^{\text{Am}} = 3.926 \neq 4.819$. The equality does not hold because the American put contains an early exercise premium. The call option does not have this premium, since it is never optimal to exercise a call early when the underlying stock does not pay dividends.

6 Question 6: Empirical Option Pricing

6.1 Data Collection

Amazon.com Inc. (AMZN) is selected as the underlying asset. AMZN does not pay dividends, satisfying the model assumption. Daily adjusted closing prices are obtained from Yahoo Finance over the one year period ending 6 March 2026 (Yahoo Finance, 2026). The most recent closing price is $S_t = 213.21$. The nearest available option maturity to nine months is 20 November 2026 (259 days, $T \approx 0.7096$ years).

6.2 Returns Distribution

Daily continuously compounded returns $r_t = \ln(S_t/S_{t-1})$ are annualised assuming 252 trading days:

$$\mu = 252\,\bar{r} = 0.0683, \qquad \sigma = \sqrt{252}\,s_r = 0.3498.$$

$$r_t \sim \mathcal{N}(\mu,\, \sigma^2)$$

6.3 Binomial Parameters

A three-period binomial tree is used to price a nine-month option. With time-to-maturity $T = 0.75$ and time step $h = T/3 = 0.25$, the Cox–Ross–Rubinstein parameterisation gives

$$u = e^{\sigma\sqrt{h}} = 1.1911, \quad d = e^{-\sigma\sqrt{h}} = 0.8395, \quad R = e^{rh} = e^{0.01}, \quad q = \frac{e^{rh} - d}{u - d} = 0.4851.$$

6.4 Three-Period Binomial Pricing of a European Put and Replicating Portfolio

The stock price at node (n, k) is $S_{n,k} = S_0\, u^k\, d^{n-k}$, and the put payoff at maturity is $V_{n,k} = \max(K - S_{n,k},\, 0)$ (where n is the period number and k is the number of up moves taken to

reach that step). Values are computed by backward induction:

$$V_t = e^{-rh}\big(q\,V_{t+1}^u + (1-q)\,V_{t+1}^d\big).$$

where e^{-rh} is the discount factor and $\big(qV_{t+1}^u + (1-q)V_{t+1}^d\big)$ is the risk-neutral expected value of the option in the next period. The option price today is therefore $P_0 = V_0$. Taking $S_0 = 213.21$ and $K = S_0 + 10 = 223.21$, the tree yields $\mathbf{P_0 = 29.42}$. Figure 2 shows all node values; Appendix B.5 provides the full backward induction.

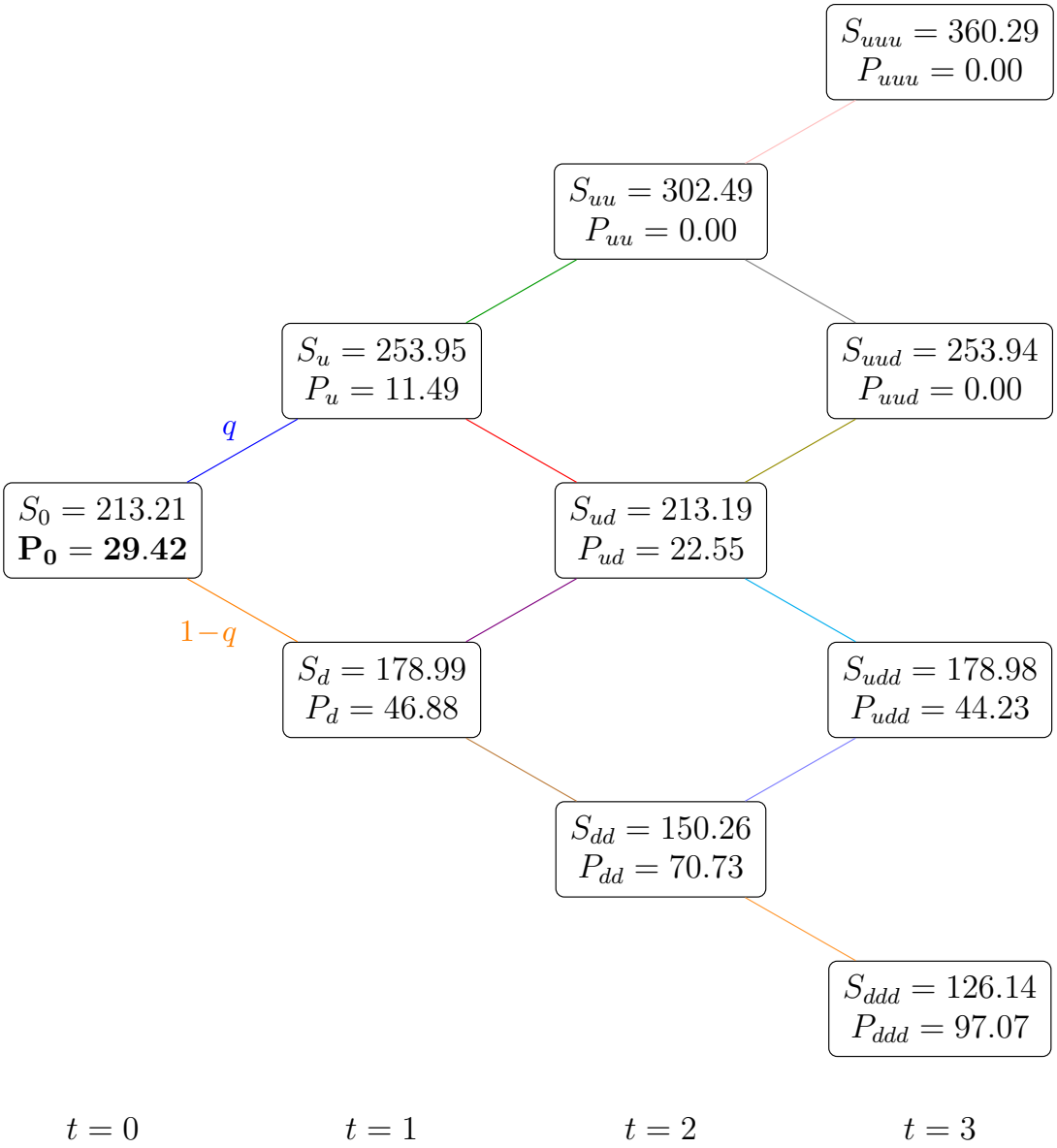

Figure 2: Three-Period Binomial Tree of Stock Price and Put Value ($K = 223.21$).

Replicating portfolio. The hedge ratio and bond position at inception are

$$\Delta_0 = \frac{P_u - P_d}{S_0(u-d)} = -0.4721, \qquad B_0 = \frac{P_d - \Delta_0 \cdot S_d}{e^{rh}} = 130.08.$$

$$P_0 = \Delta_0 S_0 + B_0 = (-0.4721)(213.21) + 130.08 = 29.42. \checkmark$$

The option price can therefore be replicated by shorting 0.4721 units of the asset and buying \$130.08 of zero-coupon bonds

Market comparison. The closest traded put (Table 3) has a mid-price of \$31.10, close to our \$29.42. The residual difference is attributable to the strike mismatch (\$225 vs. \$223.21), maturity mismatch (259 vs. 270 days), and the three-step tree's coarse terminal distribution, which limits the model to only four possible endpoint prices.

Table 3: Selected market put option data

Parameter	**Value**
Contract symbol	AMZN261120P00225000
Strike price	\$225.00
Bid / Ask	\$30.10 / \$32.10
Mid-price	\$31.10
Days to expiry	259
Time to maturity T	0.7096 years

6.5 Physical vs. Risk-Neutral Valuation

The physical up-move probability is

$$p = \tfrac{1}{2} + \tfrac{1}{2}\,\frac{\mu}{\sigma}\sqrt{h} = 0.5488 > 0.4851 = q.$$

Since $\mu > r$, Amazon has a positive equity risk premium and therefore $p > q$. This implies that under the physical measure the stock has a stronger upward drift, making the put less likely to finish in-the-money. As a result, the present value of the physical expected payoff is typically lower than the no-arbitrage price obtained from the risk-neutral binomial model. This is intuitive because option prices reflect risk-adjusted (risk-neutral) probabilities rather than the raw expected payoff under the physical distribution. Indeed,

$$P_0^{\mathbb{P}} = \$23.03 \;<\; \$29.43 = P_0^{\mathbb{Q}}.$$

6.6 Barrier Option Pricing via Monte Carlo

Monte Carlo simulation (N = 10,000 risk-neutral GBM paths, daily frequency, $\Delta t = 1/180$, seed 42) yields a KO put price of \$24.20 (95% CI [\$23.54, \$24.85]) against a vanilla put of \$28.01, a barrier cost of \$3.81. Paths evolve as

$$S_{t+\Delta t} = S_t \exp\Big(\big(r - \tfrac{1}{2}\sigma^2\big)\,\Delta t + \sigma\sqrt{\Delta t}\,Z\Big), \quad Z \sim N(0,1).$$

The vanilla price averages $e^{-rT}\max(K - S_T^{(i)}, 0)$ over all paths; the KO price zeros out the 4,989 paths (49.9%) where the stock ever exceeded $B = 1.2S_0 = \$256$:

$$\Pi_i = \begin{cases} e^{-rT}\max(K - S_T^{(i)},\,0) & \text{if } \max_t S_t^{(i)} < B, \\ 0 & \text{otherwise.} \end{cases}$$

Table 4: Path classification and price decomposition ($N = 10{,}000$)

Outcome	Paths	Share
Survived and in-the-money	4,537	45.4%
Survived and out-of-the-money	474	4.7%
Knocked out, would have been ITM	1,293	12.9%
Knocked out, would have been OTM	3,696	37.0%
KO put price ($)	24.20	
Vanilla put price ($)	28.01	
Value destroyed ($)	3.81	

The vanilla put price is estimated via Monte Carlo on the same paths, ignoring barrier status. Value destroyed equals the discounted expected payoff on the 1,293 knocked-out ITM paths.

Although 49.9% of paths knock out, the \$3.81 barrier cost is modest because 3,696 knocked-out paths (37.0%) would have expired OTM and carried no put value. Only the 1,293 paths (12.9%) that knocked out but would have paid off drive the loss. Surviving ITM paths average \$54.96; applying this to the 1,293 lost paths recovers the gap: $1{,}293 \times \$54.96/10{,}000 \times e^{-rT} \approx \$3.81$, matching $\hat{V}_{\text{vanilla}} - \hat{V}_{\text{KO}}$ exactly. The barrier destroys value by extinguishing the option on paths where a temporary breach of \$256 is followed by recovery below the strike,a trajectory-dependent loss invisible in the terminal price alone.

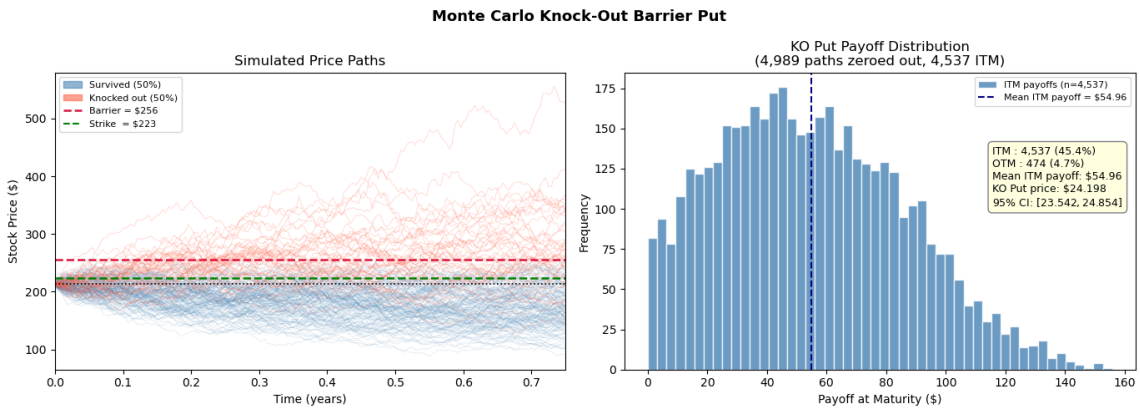

Figure 3: *Left: GBM paths coloured by survival (blue) and knock-out (red), against barrier $B = \$256$ and strike $K = \$223$. Right: payoff distribution of the 4,537 surviving ITM paths; the \$54.96 mean applied to the 1,293 lost ITM paths recovers the \$3.81 price gap.*

6.7 Variance Reduction with Control Variates

The vanilla put serves as the control variate: because payoffs are identical on every surviving path, the two estimators share most of their simulation noise. The corrected estimator is

$$\hat{V}_{\text{KO}}^{CV} = \hat{V}_{\text{KO}} - \beta\Big(\hat{V}_{\text{vanilla}} - V_{\text{BS}}\Big),$$

where $V_{\text{BS}} = \$27.61$ anchors the correction. The simulated vanilla price of \$28.01 lies within \$0.40 of V_{BS}, confirming correct model specification.

Table 5: Control variate results ($N = 10{,}000$)

Measure	Plain MC	Control Variate
KO put price ($)	24.20	23.83
95% CI ($)	[23.54, 24.85]	[23.58, 24.07]
CI width ($)	1.31	0.50
Correlation ρ	0.9253	
Coefficient β	0.9330	
Variance reduction	85.61%	

Both estimates use $N = 10{,}000$ paths with seed 42. β is the optimal OLS coefficient. CI widths are full 95% interval spans.

The CI narrows from \$1.31 to \$0.50 (62% width reduction, 85.6% variance reduction). With payoff correlation of 0.925, $\beta = 0.933$ removes nearly all shared simulation noise. The \$0.37 point-estimate shift reflects anchoring on V_{BS} rather than the simulated vanilla price,a property of the estimator, not the option. Retaining only 14.4% of original variance is equivalent to a 7× sample-size increase; the CV estimator thus matches the precision of 70,000 plain MC paths at negligible extra cost.

A Group Member Contributions

Vatsal completed Questions 1, 2, and 4. Gokul completed Questions 3 and 5. Question 6 was completed jointly by Lucas, Murray, and Elvio.

B Detailed Calculations

This appendix provides the full step by step derivations underlying the results reported in the main text.

B.1 Question 2: Put–Call Parity Arithmetic

Present value of strike price.

$$Ke^{-rT} = 235 \times e^{-0.01\times 90/365} = 235 \times e^{-0.002466} = 235 \times 0.997537 = 234.4213.$$

Parity check.

$$\text{LHS: } C_0 + Ke^{-rT} = 21.88 + 234.4213 = 256.3013.$$

$$\text{RHS: } P_0 + S_0 = 5.25 + 250 = 255.25.$$

$$\text{Difference: } 256.3013 - 255.25 = 1.0513.$$

Parity-implied put.

$$P_0^* = C_0 + Ke^{-rT} - S_0 = 21.88 + 234.4213 - 250 = 6.3013.$$

$$\text{Mispricing: } P_0^* - P_0 = 6.3013 - 5.25 = 1.0513.$$

Net cash flow at $t = 0$.

$$-S_0 - P_0 + C_0 + Ke^{-rT} = -250 - 5.25 + 21.88 + 234.4213 = +1.0513.$$

Terminal payoff verification.

If $S_T \geq K$:

$$S_T + \max(K - S_T, 0) - \max(S_T - K, 0) - K = S_T + 0 - (S_T - K) - K = 0.$$

If $S_T < K$:

$$S_T + (K - S_T) - 0 - K = S_T + K - S_T - K = 0.$$

B.2 Question 3: Brownian Motion Expansion

Let $Z := W_{t+2} - W_t \sim N(0, 2)$, independent of $\mathcal{F}_t$. Then $W_{t+2} = W_t + Z$.

Moments of Z. Since $Z \sim N(0, 2)$: $\mathbb{E}[Z] = 0$, $\mathbb{E}[Z^2] = \mathrm{Var}(Z) = 2$, $\mathbb{E}[Z^3] = 0$ (all odd moments of a centred Gaussian vanish).

First term.

$$\mathbb{E}_t[W_{t+2}] = \mathbb{E}_t[W_t + Z] = W_t + \mathbb{E}[Z] = W_t.$$

Second term.

$$\begin{aligned}
\mathbb{E}_t[W_{t+2}^2] &= \mathbb{E}_t[(W_t + Z)^2] \\
&= \mathbb{E}_t[W_t^2 + 2W_t Z + Z^2] \\
&= W_t^2 + 2W_t \cdot \mathbb{E}[Z] + \mathbb{E}[Z^2] \\
&= W_t^2 + 0 + 2 = W_t^2 + 2.
\end{aligned}$$

Third term.

$$\begin{aligned}
\mathbb{E}_t[W_{t+2}^3] &= \mathbb{E}_t[(W_t + Z)^3] \\
&= \mathbb{E}_t[W_t^3 + 3W_t^2 Z + 3W_t Z^2 + Z^3] \\
&= W_t^3 + 3W_t^2 \cdot \mathbb{E}[Z] + 3W_t \cdot \mathbb{E}[Z^2] + \mathbb{E}[Z^3] \\
&= W_t^3 + 0 + 6W_t + 0 = W_t^3 + 6W_t.
\end{aligned}$$

Combining.

$$\begin{aligned}
\mathbb{E}_t\big[W_{t+2} + 2W_{t+2}^2 + 3W_{t+2}^3\big] &= W_t + 2(W_t^2 + 2) + 3(W_t^3 + 6W_t) \\
&= W_t + 2W_t^2 + 4 + 3W_t^3 + 18W_t \\
&= 3W_t^3 + 2W_t^2 + 19W_t + 4.
\end{aligned}$$

B.3 Question 4: One-Period Binomial Derivations

Risk-neutral probability. From $S_0R = qS_u + (1-q)S_d$:

$$110 \times 1.05 = q(121) + (1-q)(99) \implies 115.5 = 99 + 22q \implies q = \frac{16.5}{22} = 0.75.$$

Option payoffs.

$$C_u = \max(121 - 100, 0) = 21, \quad C_d = \max(99 - 100, 0) = 0.$$

$$P_u = \max(100 - 121, 0) = 0, \quad P_d = \max(100 - 99, 0) = 1.$$

Risk-neutral pricing.

$$C_0 = \frac{0.75(21) + 0.25(0)}{1.05} = \frac{15.75}{1.05} = 15.$$

$$P_0 = \frac{0.75(0) + 0.25(1)}{1.05} = \frac{0.25}{1.05} = 0.238095.$$

Put replication. Solve $S_u\Delta + RB = P_u$ and $S_d\Delta + RB = P_d$:

$$121\Delta + 1.05B = 0, \qquad 99\Delta + 1.05B = 1.$$

Subtracting: $22\Delta = -1$, so $\Delta = -1/22$. Then $1.05B = -121(-1/22) = 5.5$, so $B = 5.5/1.05 = 5.2381$.

Verification: $\Delta S_0 + B = (-1/22)(110) + 5.2381 = -5 + 5.2381 = 0.2381 = P_0$. ✓

Call replication. Solve:

$$121\Delta + 1.05B = 21, \qquad 99\Delta + 1.05B = 0.$$

Subtracting: $22\Delta = 21$, so $\Delta = 21/22$. Then $1.05B = -99(21/22) = -94.5$, so $B = -94.5/1.05 = -90$.

Verification: $\Delta S_0 + B = (21/22)(110) - 90 = 105 - 90 = 15 = C_0$. ✓

Put–call parity.

$$\text{LHS: } C_0 + K/R = 15 + 100/1.05 = 15 + 95.2381 = 110.2381.$$

$$\text{RHS: } P_0 + S_0 = 0.2381 + 110 = 110.2381. \checkmark$$

B.4 Question 5: Two-Period American Option Derivations

Parameters. $S_0 = 100$, $K = 100$, $u = 1.10$, $d = 0.90$, $R = 1.025$, $q = 0.625$.

Stock lattice.

$$S_u = 100(1.10) = 110, \quad S_d = 100(0.90) = 90.$$
$$S_{uu} = 110(1.10) = 121, \quad S_{ud} = 110(0.90) = 99, \quad S_{dd} = 90(0.90) = 81.$$

American call.

Terminal payoffs: $C_{uu} = (121-100)^+ = 21$, $C_{ud} = (99-100)^+ = 0$, $C_{dd} = (81-100)^+ = 0$.

At node u:

$$C_u^{\text{cont}} = \frac{q \cdot C_{uu} + (1-q) \cdot C_{ud}}{R} = \frac{0.625(21) + 0.375(0)}{1.025} = \frac{13.125}{1.025} = 12.8049.$$

Exercise value: $(110-100)^+ = 10 < 12.8049$. Hence $C_u = 12.8049$.

At node d:

$$C_d^{\text{cont}} = \frac{0.625(0) + 0.375(0)}{1.025} = 0.$$

Exercise value: $(90-100)^+ = 0$. Hence $C_d = 0$.

At $t = 0$:

$$C_0 = \frac{0.625(12.8049) + 0.375(0)}{1.025} = \frac{8.0030}{1.025} = 7.8078.$$

Exercise value: $(100-100)^+ = 0 < 7.8078$. Hence $C_0^{\text{Am}} = C_0^{\text{Eu}} = 7.808$.

American put.

Terminal payoffs: $P_{uu} = (100-121)^+ = 0$, $P_{ud} = (100-99)^+ = 1$, $P_{dd} = (100-81)^+ = 19$.

At node u:

$$P_u^{\text{cont}} = \frac{0.625(0) + 0.375(1)}{1.025} = \frac{0.375}{1.025} = 0.3659.$$

Exercise value: $(100-110)^+ = 0 < 0.3659$. Hence $P_u^{\text{Am}} = 0.3659$.

At node d:

$$P_d^{\text{cont}} = \frac{0.625(1) + 0.375(19)}{1.025} = \frac{0.625 + 7.125}{1.025} = \frac{7.75}{1.025} = 7.5610.$$

Exercise value: $(100-90)^+ = 10 > 7.5610$. **Early exercise is optimal.** Hence $P_d^{\text{Am}} = 10$.

European put at $t = 0$:

$$P_0^{\text{Eu}} = \frac{0.625(0.3659) + 0.375(7.5610)}{1.025} = \frac{0.2287 + 2.8354}{1.025} = \frac{3.0641}{1.025} = 2.9894.$$

American put at $t = 0$:

$$P_0^{\text{Am}} = \frac{0.625(0.3659) + 0.375(10)}{1.025} = \frac{0.2287 + 3.75}{1.025} = \frac{3.9787}{1.025} = 3.8817.$$

Put–call parity check.

$$R^2 = 1.025^2 = 1.050625, \qquad K/R^2 = 100/1.050625 = 95.1814.$$

$$S_0 - K/R^2 = 100 - 95.1814 = 4.8186.$$

$$C_0^{\text{Eu}} - P_0^{\text{Eu}} = 7.8078 - 2.9894 = 4.8184 \approx 4.8186. \checkmark$$

$$C_0^{\text{Am}} - P_0^{\text{Am}} = 7.8078 - 3.8817 = 3.9261 \neq 4.8186.$$

B.5 Question 6b: Backward Induction and Replicating Portfolio

Parameters. $S_0 = 213.21$, $K = 223.21$, $u = 1.1911$, $d = 0.8395$, $R = e^{0.01} = 1.01005$, $q = 0.4851$.

Stock prices at each node.

$$S_u = 213.21 \times 1.1911 = 253.95, \qquad S_d = 213.21 \times 0.8395 = 178.99.$$
$$S_{uu} = 253.95 \times 1.1911 = 302.49, \qquad S_{ud} = 253.95 \times 0.8395 = 213.19.$$
$$S_{dd} = 178.99 \times 0.8395 = 150.26.$$

$$S_{uuu} = 302.49 \times 1.1911 = 360.29, \qquad S_{uud} = 302.49 \times 0.8395 = 253.94.$$
$$S_{udd} = 213.19 \times 0.8395 = 178.98, \qquad S_{ddd} = 150.26 \times 0.8395 = 126.14.$$

Terminal put payoffs ($t = 3$).

$$\begin{aligned}
P_{uuu} &= \max(223.21 - 360.29,\ 0) = 0.00,\\
P_{uud} &= \max(223.21 - 253.94,\ 0) = 0.00,\\
P_{udd} &= \max(223.21 - 178.98,\ 0) = 44.23,\\
P_{ddd} &= \max(223.21 - 126.14,\ 0) = 97.07.
\end{aligned}$$

Backward induction ($t = 2$).

$$\begin{aligned}
P_{uu} &= e^{-0.01}\big[0.4851(0) + 0.5149(0)\big] = 0.00.\\
P_{ud} &= e^{-0.01}\big[0.4851(0) + 0.5149(44.23)\big] = e^{-0.01}(22.78) = 22.55.\\
P_{dd} &= e^{-0.01}\big[0.4851(44.23) + 0.5149(97.07)\big] = e^{-0.01}(71.44) = 70.73.
\end{aligned}$$

Backward induction ($t = 1$).

$$\begin{aligned}
P_u &= e^{-0.01}\big[0.4851(0) + 0.5149(22.55)\big] = e^{-0.01}(11.61) = 11.49.\\
P_d &= e^{-0.01}\big[0.4851(22.55) + 0.5149(70.73)\big] = e^{-0.01}(47.36) = 46.88.
\end{aligned}$$

Price at $t = 0$.

$$P_0 = e^{-0.01}\big[0.4851(11.49) + 0.5149(46.88)\big] = e^{-0.01}(29.72) = 29.42.$$

Replicating portfolio derivation.

The hedge ratio at $t = 0$:

$$\Delta_0 = \frac{P_u - P_d}{S_u - S_d} = \frac{11.49 - 46.88}{253.95 - 178.99} = \frac{-35.39}{74.96} = -0.4721.$$

The bond position:

$$B_0 = \frac{P_d - \Delta_0 \cdot S_d}{e^{rh}} = \frac{46.88 - (-0.4721)(178.99)}{e^{0.01}} = \frac{46.88 + 84.50}{1.01005} = \frac{131.38}{1.01005} = 130.08.$$

Verification:

$$\Delta_0 S_0 + B_0 = (-0.4721)(213.21) + 130.08 = -100.66 + 130.08 = 29.42. \checkmark$$

B.6 Question 6c: Physical Probability Calculation

$$p = \frac{1}{2} + \frac{1}{2}\frac{\mu}{\sigma}\sqrt{h} = \frac{1}{2} + \frac{1}{2}\frac{0.0683}{0.3498}\sqrt{0.25} = 0.5 + 0.5 \times 0.1953 \times 0.5 = 0.5488.$$

Physical expected payoff. Using p in place of q in the binomial tree:

$$\mathbb{E}^{\mathbb{P}}[\text{payoff}] = p^3 P_{uuu} + 3p^2(1-p)P_{uud} + 3p(1-p)^2 P_{udd} + (1-p)^3 P_{ddd}.$$

Discounting at the risk-free rate:

$$P_0^{\mathbb{P}} = e^{-rT}\,\mathbb{E}^{\mathbb{P}}[\text{payoff}] = \$23.03.$$

C Python Code

The Python notebook used for the analysis can be accessed via the following link:

Google Colab Notebook

The full link is also provided below:

`https://colab.research.google.com/drive/1YioV-35R-H0ejD8Vi4v5eSFyYIQne32R?usp=sharing`

References

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Yahoo Finance (2026). Amazon.com, Inc. (AMZN) Historical Prices. Available at: https://finance.yahoo.com/quote/AMZN/history (Accessed: 6 March 2026).